

range

```
In [1]: range(5)
```

```
Out[1]: range(0, 5)
```

```
In [2]: list(range(0,5))
```

```
Out[2]: [0, 1, 2, 3, 4]
```

```
In [3]: list(range(10,20))
```

```
Out[3]: [10, 11, 12, 13, 14, 15, 16, 17, 18, 19]
```

```
In [5]: list(range(10,20,3))
```

```
Out[5]: [10, 13, 16, 19]
```

```
In [7]: list(range(10,20,5,3))
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[7], line 1  
----> 1 list(range(10,20,5,3))  
  
TypeError: range expected at most 3 arguments, got 4
```

```
In [8]: r = range(0,11,3)  
r
```

```
Out[8]: range(0, 11, 3)
```

```
In [9]: for i in r:  
        print(i)
```

```
0  
3  
6  
9
```

```
In [10]: list(r)
```

```
Out[10]: [0, 3, 6, 9]
```

advance slicing

```
In [12]: l = ['a', 'b', 'c', 1, 2.5, 62, True, 1+2j]  
l
```

```
Out[12]: ['a', 'b', 'c', 1, 2.5, 62, True, (1+2j)]
```

```
Out[13]: ['a', 'b', 'c', 1, 2.5, 62, True, (1+2j)]
```

```
In [14]: l[0:]
```

```
Out[14]: ['a', 'b', 'c', 1, 2.5, 62, True, (1+2j)]
```

```
In [15]: l[3:]
```

```
Out[15]: [1, 2.5, 62, True, (1+2j)]
```

```
In [16]: l[-5:]
```

```
Out[16]: [1, 2.5, 62, True, (1+2j)]
```

```
In [17]: l[3]
```

```
Out[17]: 1
```

```
In [18]: l[-5]
```

```
Out[18]: 1
```

```
In [19]: l[2:9]
```

```
Out[19]: ['c', 1, 2.5, 62, True, (1+2j)]
```

```
In [20]: l[2:7]
```

```
Out[20]: ['c', 1, 2.5, 62, True]
```

```
In [22]: l[2:7:2]
```

```
Out[22]: ['c', 2.5, True]
```

```
In [23]: l[0:7:5]
```

```
Out[23]: ['a', 62]
```

```
In [25]: l[-5:3] #slicing cannot start backwards
```

```
Out[25]: []
```

```
In [26]: l[-5:-3]
```

```
Out[26]: [1, 2.5]
```

```
In [28]: s=['sbi','icici','hdfc','central','union','canara','city']  
s
```

```
Out[28]: ['sbi', 'icici', 'hdfc', 'central', 'union', 'canara', 'city']
```

```
In [29]: len(s)
```

```
Out[29]: 7
```

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [30]: `s[6]`

Out[30]: `'city'`

In [31]: `s[-1]`

Out[31]: `'city'`

In [33]: `s[-7:-1]`

Out[33]: `['sbi', 'icici', 'hdfc', 'central', 'union', 'canara']`

In [34]: `s[::-1]`

Out[34]: `['city', 'canara', 'union', 'central', 'hdfc', 'icici', 'sbi']`

In [35]: `s[::-2]`

Out[35]: `['city', 'union', 'hdfc', 'sbi']`

In [36]: `s[::-5]`

Out[36]: `['city', 'icici']`

In [38]: `2^6`

Out[38]: `4`

In []: